

Database

Database is a way for organizing data in a structured way where the user can retrieve, insert, store and manage data in an effective way. The core of a database system is a database management system which acts as an intermediary between users and the raw data. The **DBMS** handles tasks like querying, updating, deleting and managing access permissions, without requiring users to know the physical details of where data is stored. There are many types of databases; each type is classified based on a way of organizing data.

Relationship Database: is a way for organizing data in a table that is made from rows and columns. They use schemas (blueprints) to define how data is structured and how different tables relate to each other. There are many software's that use a relationship database to store data such as : MySQL, PostgreSQL, Oracle, Microsoft SQL Server.

NoSql Database: it is created for storing unstructured data such as text, image and videos. It stores data as Document Databases as JSON format, Key Value store which store the key that map to its value, columnar database which store data as columns that are very useful for analysis, and graph database which store data as a node and relationship between nodes.

Object-Orientated database: it stores the data as an object as we know each object has attribute and function therefore it is very efficient for class for implementing inheritance and encapsulation and can work even if we have complex data instead of storing data and converting it into a table can store the object directly.

Hierarchical Database: it stores the data in a tree relationship as parent and child it is very effective fast accessing for data.

Types of Databases

Relational Database



Stores data in tables

NoSQL Database



Unstructured data

Object-Oriented Database



Object storage

Hierarchical Database



Tree-like structure

Three-Tier Database Architecture

Made with AI

Client Tier



User Interface

Responses

Application Tier



Application Server



Database Tier



DBMS

Data base management system is a software used to store, organize, retrieve and manage data. The main technologies that use DBMS are oracle Database, Microsoft SQL Server, PostgreSQL, and MySQL. Although all four support SQL and relational data such as Oracle, Microsoft SQL Server, PostgreSQL, and MySQL are all relational database management systems (RDBMS). They perform the same basic tasks: Store data, Retrieve data, Insert and update data, Delete data, Manage transactions, Protect data, Provide backup and recovery.

However, they differ in scalability, performance, security, licensing, enterprise support, cloud readiness, backup/recovery, and integration with other technologies.

1.Oracle

Oracle is designed for very large and mission-critical workloads.

It supports advanced enterprise scalability technologies such as Oracle Real Application Clusters (RAC) and engineered systems such as Exadata. Oracle's current licensing documentation identifies RAC as an Enterprise Edition scalability option.

This makes Oracle particularly suitable when an organization needs:

Very large databases

Large numbers of simultaneous users

High transaction volumes

High availability

Large enterprise applications

Performance

Oracle is particularly strong for:

High-volume transaction processing

Complex queries

Large databases

Data warehousing

Enterprise applications

Oracle's Enterprise Edition is specifically positioned for high-volume transaction processing and query-intensive workloads.

Security

Oracle provides extensive enterprise security capabilities, including encryption and security controls.

2. Microsoft SQL Server

SQL Server is designed to scale from relatively small applications to large enterprise systems.

Microsoft describes SQL Server Enterprise as intended for mission-critical performance, security, and high availability, while Standard is targeted at smaller/mid-sized workloads.

SQL Server provides technologies for: High availability, Failover, Replication, Large databases,, Enterprise workloads, Cloud deployment.

Performance.

SQL Server is strong for:

Transaction processing

Business applications

Reporting

Analytics

Data warehousing

Security

SQL Server provides extensive security capabilities, including:

Authentication

Authorization

Encryption

Auditing

Row-level security

Threat monitoring

3. PostgreSQL

PostgreSQL is an open-source object-relational DBMS. It is known for reliability, data integrity, extensibility, SQL support, and its ability to handle complex workloads.

Unlike Oracle and SQL Server, PostgreSQL is developed as an open-source project with a broad community and vendor ecosystem.

PostgreSQL can support small applications as well as large enterprise workloads.

Its scalability can be achieved through: Replication, Partitioning, Connection pooling, Read replicas, Distributed architectures, Cloud infrastructure

, highly extensible.

Performance

PostgreSQL is particularly strong for:

Complex SQL

Complex queries

Transaction processing

Data integrity

JSON workloads

Geographic data

Analytical workloads

PostgreSQL also supports extensions such as PostGIS for geographic information.

Security

PostgreSQL provides:

Authentication

Authorization

Roles

Privileges

SSL/TLS

Row-level security

Encryption capabilities

Auditing through extensions/tools

It provides a strong security foundation without requiring organizations to purchase a proprietary enterprise database license.

4. MySQL

MySQL is a widely used relational DBMS with a particularly strong history in web applications and online services.

It is used by organizations across technology, financial services, telecommunications, manufacturing, media, education, and e-commerce. MySQL supports: Replication, Clustering, Read replicas, InnoDB Cluster, High availability, Cloud deployments.

MySQL states that it is used for demanding Web, E-commerce, SaaS, and OLTP applications.

Performance

MySQL is particularly well known for:

Web applications

Online transaction processing

E-commerce

SaaS applications

High-volume application workloads

Its InnoDB storage engine provides transaction safety, ACID compliance, crash recovery, and row-level locking.

Security

MySQL provides:

Authentication

Authorization

Encryption

Roles

Auditing

Enterprise security tools

Commercial MySQL Enterprise Edition adds additional security, monitoring, backup, and management capabilities.

main difference among DBMS technologies

ORACLE	SQL Server	PostgreSQL	MySQL
Mission-Critical Systems	Business & Analytics	Complex Data & Open Source	Web & CMS Platforms
• Banking & Financial Services • Telecom Infrastructure • Enterprise Resource Planning (ERP)	• Corporate Applications • Business Intelligence (BI) • Reporting & Dashboards	• Geospatial & Research Data • Data Warehousing • Open Source Projects	• E-Commerce Websites • Content Management (CMS) • LAMP Stack Applications
			
 High Security	 BI & Analytics	 Geospatial Data	 Web & CMS